

BEYOND AVERAGING: ALL-ROUNDER CAPABILITY EVALUATION FOR RADIOLOGY REPORT GENERATION

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Introduction

Radiology Report Generation (RRG) aims to automatically generate clinically accurate reports from medical images, improving efficiency in clinical workflows.

However, current evaluation practices suffer from an evaluation bottleneck. Existing metrics (e.g., BLEU, ROUGE-L, CheXbert F1) compute a single aggregate score, which masks performance disparities caused by data imbalance. As a result, models may perform well on frequent cases while failing on rare but clinically important scenarios.

To address this limitation, we propose All-Rounder Capability Evaluation (ARCE), a subgroup-based evaluation framework that explicitly measures both performance and fairness across clinically meaningful subsets. By combining subgroup mean performance with a Gini-based fairness metric, ARCE provides a more reliable assessment of model generalization. Our contributions:

1. Proposing first subgroup-based evaluation framework for RRG
2. Introducing ARCE metric which balances mean performance and fairness across subgroups to quantify generalization ability.
3. Demonstrating ARCE's flexibility and generalizability by applying it to other different subgroup criteria.

Motivation

Current evaluation metrics fail to reflect real-world model reliability.

1. **Data Imbalance:** RRG datasets are inherently imbalanced across subgroups (e.g., imaging views, disease rarity).
2. **Misleading Rankings:** Standard metrics (e.g., BLEU, CheXpert F1) are all Macro Aggregate Metrics (MAM) from simple averaging perspective over entire test set, **this masks poor generalization on underrepresented cases.**
3. **Fairness Gap:** Existing metrics do not capture fairness across different clinical scenarios.

Method

Clinically Meaningful Subgroups Definition:

- CF (Current Frontal): Contains only the current anteroposterior view, reflecting basic diagnostic capabilities.
- CF + CL (Current Lateral): Lateral views provide information about lesions that may be obscured in anteroposterior views, aiding in lesion localization.
- CF + PF (Prior Frontal): Adding historical frontal views for longitudinal comparison to assess disease progression or improvement.
- CF + PF + CL (Comprehensive): The scenario with the richest information and the highest diagnostic challenge, requiring the integration of multiple views and time-series information.

The cornerstone of our evaluation paradigm is the stratification of the test set into clinically meaningful, mutually exclusive subgroups.

The partitioning chosen for this main experiment is based on the types of imaging views included in each radiology examination, which directly reflects the diversity of clinical workflows and the richness of diagnostic information.

Method

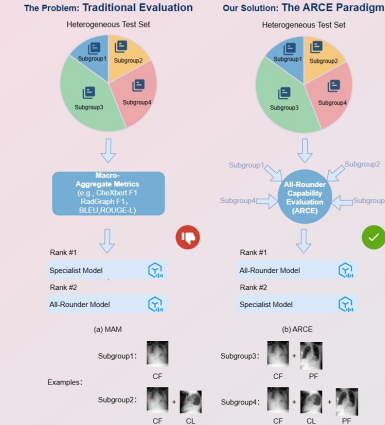


Fig. 1. Compared to MAM, our ARCE can obviously capture the imbalance gap by discriminating different subgroups.

Why Gini instead of standard deviation ?

- Measures fairness rather than statistical dispersion
- Scale-invariant → comparable across metrics
- Captures subgroup-wise imbalance via pairwise differences
- Less sensitive to outliers → reflects systematic disparity

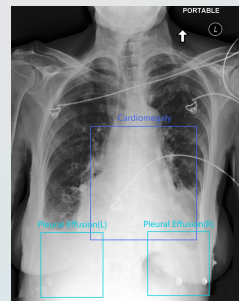
Hidden Failure Case

Subgroup scores [CF, CFCL, CFFP, ALL]: [0.4, 0.35, 0.88, 0.9]

MAM: Avg = 0.63 (misleading: looks good but ✗) ARCE: Mean = 0.63, Gini is **High** → Penalized score ↓

Macro Aggregate average hides poor performance on underrepresented subgroups.

Results



Ground Truth: Small bilateral pleural effusions are present. There is no focal consolidation. The lungs are hyperinflated with emphysematous changes. Heart size is increased.

METRIC₁: The heart size is top normal in configuration, unchanged since the prior study. There are diffuse patchy opacities bilaterally, concerning for pneumonia or aspiration. Mild pulmonary edema is also noted. No large pleural effusion seen.

METRIC₂: The right hemidiaphragm is elevated with air under the apex. There are multiple areas of consolidation in the left lung, including bilateral pleural effusions and possible pulmonary edema. No pneumothorax or pneumomediastinum. Heart size is enlarged.

Fig. 2. Case study on the CF subgroup. Image: input frontal chest X-ray. Text: generated reports from different models compared with the ground truth. The highlighted part is the lesion reflected in this paragraph of text. Blue is correct and red is wrong.

The ARCE Framework

The All-Rounder Capability Evaluation (ARCE) score is defined as:

$$ARCE = \mu \cdot (1 - G)$$

μ : the mean performance across subgroups

G : the Gini coefficient of the subgroup scores.

Formally, μ is defined as:

$$\mu = \frac{\sum_{i=0}^N w_i s_i}{\sum_{i=0}^N w_i}$$

w_i : the size (or relative importance) of subgroup D_i .

s_i : subgroup-specific scores for a chosen MAM metric (e.g., CheXbert F1, RadGraph F1).

In our research, we set all weights $w_i = 1$ to treat each subgroup equally. Intuitively, μ reflects the overall ability of the model. This design ensures that ARCE does not only emphasize fairness but also rewards strong overall performance.

The calculations for G is:

$$G = \frac{\sum_{i=0}^N \sum_{j=0}^N w_i s_i |s_i - s_j|}{2\mu \sum_{i=0}^N \sum_{j=0}^N w_i s_i}$$

G measures performance inequality across subgroups. Lower G → more consistent and fair performance. Higher G → performance imbalance.

G penalizes models that perform well only on dominant cases and prevents these models from being overvalued.

Results

Table 1. Baseline Performance with traditional MAM Evaluation.

MAM	BLEU1	ROUGE L	neg Macro 14 F1	neg Micro 14 F1
Model ₁	27.47	22.03	28.15	45.15
Model ₂	27.22	20.86	27.23	43.34
Model ₃	27.42	19.61	27.18	43.68

Table 2. Subgroup Analysis of Model Performance

Macro/Micro 14	Model ₁	Model ₂	Model ₃
CF	20.13/35.54	22.77/39.28	23.06/39.75
CF+CL	20.99/33.23	21.04/33.14	21.06/30.85
CF+PF	29.73/50.90	27.25/47.87	27.92/48.49
CF+PF+CL	25.78/38.97	27.29/38.76	26.48/38.43

Table 3. Evaluation with Our Proposed Strategy ARCE

ARCE	BLEU1	ROUGE L	neg Macro 14 F1	neg Micro 14 F1
Model ₁	26.58	21.37	22.06	36.13
Model ₂	26.58	20.47	23.13	36.97
Model ₃	27.04	19.46	23.13	35.99

Table 4. Validation on a Balanced Test Set

ARCE	BLEU1	ROUGE L	neg Macro 14 F1	neg Micro 14 F1
Model ₁	19.84	17.79	12.01	19.83
Model ₂	21.48	17.44	13.35	23.02
Model ₃	21.69	17.00	12.53	21.59

MAM can mis-rank models: Table 1 suggests the performance of three models on MAM. From these results, one might conclude that *Model₁* performs best, and that *Model₂* and *Model₃* are difficult to distinguish.

Subgroup analysis reveals hidden specialization: Table 2 shows that Models perform inconsistently across clinically meaningful subgroups, exposing disparities masked by MAM.

ARCE better reflects true generalization: Table 3 presents that after fairness-aware adjustment, *Model₁* is downgraded, while balanced models are favored.

ARCE aligns with balanced-set evaluation: We constructed a subgroup-balanced subset from unused MIMIC data and re-evaluated all models. Table 4 shows that the ranking derived from ARCE aligns closely with that on the balanced set.

Conclusion & Contact

Conclusion:

In this work, we reveal a fundamental limitation in current RRG evaluation: an over-reliance on MAM, which can systematically favor “specialized” models that perform well on frequent, simple cases but fail in rare and complex ones. To address this, we propose a clinically informed subgroup evaluation paradigm and introduce the All-Rounder Capability Evaluation, which jointly accounts for overall performance and subgroup fairness.

Overall, this work provides not only a practical evaluation tool but also a shift in perspective: moving from maximizing a single score toward developing AI systems that are robust, fair, and clinically trustworthy.

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